

The Endpoint Geometry of Common-Digit Fabius Laws

Sharp joint small deviations, copula universality,
continuum sampling, and noncommuting Gaussian limits

The central mechanism

Shared digits do not force shared extremes. A joint lower-tail event charges one cost for each digit, determined by the strongest active parameter constraint. The resulting upper envelope yields an explicit quadratic rate, a block-simplex correction, and exact residual tail-dependence exponents.

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29 September 2026

Status. A research manuscript with complete written proofs and reproducible finite checks. No Lean or Rocq formalization is claimed. The novelty claim is limited to the contribution developed here relative to the inspected repository report; absolute historical priority has not been established.

Repository snapshot. `e1afd75e35a4de734d5aa47aec5cdb917b82ed3e`

Source area. `Analysis/FabiusFunction`; common-digit geometric-uniform laws and the report's open endpoint-dependence program.

Abstract

Let $U_0, U_1, \dots$ be independent uniform random variables and couple all parameters through the same digits, $X_q = (1 - q) \sum_{n \geq 0} q^n U_n$. At $q = 1/2$ the marginal distribution function is the Fabius function. The existing ProvelT common-digit report determines support geometry and a Gaussianization limit, but leaves endpoint dependence as a research direction. We derive a joint small-deviation theory whose governing object is the upper envelope $h_a(x) = \max(0, a_i - \lambda_i x : 1 \leq i \leq d)$, where $q_i = e^{-\lambda_i}$. In a fixed nondegenerate envelope chamber,

$$\log \mathbb{P}(X_{q_i} \leq e^{-a_i t} \forall i) = -I(a)t^2 - T(a)t \log t + A(a)t + O((\log t)^2),$$

with all coefficients explicit. The proof uses independent simplex blocks and small transition strips rather than an unproved saddle-point inversion. For digits with density $cz^{\beta-1}(1 + O(z^\delta))$ at zero, we obtain the corresponding sharp expansion and show that marginal standardization removes c , all weight normalizations, and even β from the leading copula rate. A log-digit log-concavity argument proves actual regular variation along power rays, not merely a logarithmic equivalent.

For $A > B > 0$ and uniform digits, the diagonal copula exponent is $\kappa = 2\sqrt{A}/(\sqrt{A} + \sqrt{B})$; hence the Ledford–Tawn coefficient is $\eta = (1 + \sqrt{B/A})/2$. We calculate the first stretched-exponential correction, the complete bivariate phase diagram, and a finite-dimensional formula that becomes $1 + \frac{1}{4} \log(A/B)$ under a continuum of parameter constraints. Equally spaced logarithms of the λ_i uniquely maximize the finite-sample exponent. Finally, common hyperbolic parameter translation keeps Pearson correlation exactly constant and gives a Gaussian limit, yet its Gaussian and extreme-tail limits do not commute. We give exact finite probability certificates, reproducible checks, limitations, and a prioritized research agenda.

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1 The question, the contribution, and its boundary

1.1 What is inherited and what is addressed

The classical Fabius construction provides a compactly supported smooth probability law through an infinite series of independent uniform random variables [2, 3]. The ProveIt report *Common-Digit Fabius Zonoids* [1] couples the geometric-uniform laws at distinct parameters with the *same* digit sequence. It develops their support zonotopes, covariance geometry, cumulants, parameter jets, and a Gaussian limit under hyperbolic translation. Its section on inverse-Fabius copula endpoint laws proposes boundary transseries and conditional extreme-event asymptotics.

This manuscript studies the missing *probability mass near the common lower corner*. Exact support boundaries alone do not determine that mass. Two random variables can have a smooth joint density, very strong positive correlation, and a highly constrained common support, while simultaneous extreme quantiles are nevertheless asymptotically independent. We quantify that phenomenon and explain its mechanism.

The inspected source is pinned to the repository snapshot on the title page. Its explicit conjecture `conj:copula-endpoints` concerns *boundary curves* and their transseries. We do not claim to prove that entire conjecture. Our theorems resolve the associated joint lower-tail rate problem and its first correction; conditioning on a threshold event is also covered. Conditioning on an exact coordinate value, complete boundary transseries, and bounded periodic terms remain separate questions.

1.2 A map of the results

Result	Content and scope
Envelope law	Finite and compact-continuum joint lower-tail rates; a direct independent-box proof.
Sharp expansion	Explicit t^2 , $t \log t$, and t terms with $O((\log t)^2)$ remainder in fixed strict chambers.
Copula universality	Endpoint-index-independent leading exponent and an explicit $\sqrt{\log(1/u)}$ correction.
Regular variation	Genuine power-ray regular variation when the log-digit density is log-concave, including uniform and beta-power digits.
Parameter sampling	Closed formulas for any finite diagonal, its continuum limit, and optimal finite placement.
Gaussian comparison	Constant-correlation Gaussianization does not commute with extreme-quantile asymptotics.

The elementary simplex integral, Prékopa's marginal theorem, regular variation terminology, and the classical Gaussian tail calculation are not new. They are identified where used. The proposed contribution is the explicit envelope/simplex theory and its consequences for this common-digit family. A focused literature search did not establish an earlier identical result, but that is not a certification of historical priority.

1.3 Conventions and the interpretation of errors

All logarithms are natural. Parameters and dimension are fixed unless a limit is explicitly stated. An O -constant may depend on those parameters and on the digit law. Uniform error claims are made only on compact subsets of a specified strict chamber. In particular, no uniformity as two distinct parameters coalesce is implicit. Every asymptotic statement is a written mathematical theorem of this manuscript; the accompanying program checks finite identities and certificates, not infinite limiting assertions.

2 The model and the geometry of an upper envelope

Let $V_0, V_1, \dots$ be independent identically distributed random variables in $[0, 1]$. Set

$$q_i = e^{-\lambda_i}, \quad w_i = 1 - e^{-\lambda_i}, \quad X_i = w_i \sum_{n=0}^{\infty} e^{-\lambda_i n} V_n, \quad \lambda_i > 0. \quad (2.1)$$

The series converges absolutely and $0 \leq X_i \leq 1$. The uniform case is $V_n = U_n \sim \text{Unif}[0, 1]$. We assume the λ_i are distinct; repeated coordinates can be consolidated by retaining the most restrictive threshold.

For $a = (a_1, \dots, a_d) \in (0, \infty)^d$ define

$$E_t(a) = \{X_i \leq e^{-a_i t} : 1 \leq i \leq d\}, \quad (2.2)$$

$$h_a(x) = \max\{0, a_1 - \lambda_1 x, \dots, a_d - \lambda_d x\}, \quad (2.3)$$

$$I(a) = \int_0^\infty h_a(x) dx, \quad T(a) = \max_i \frac{a_i}{\lambda_i}. \quad (2.4)$$

For each i , let $\ell_i(a)$ be the length of the interval on which the i th line is the unique positive maximizer. Empty intervals have length zero. Ties occur at finitely many abscissae and do not affect lengths. The positive envelope occupies $[0, T(a))$.

Definition 2.1 (Strict envelope chamber). A parameter vector a lies in a strict chamber when the list of active lines is unchanged by sufficiently small perturbations of a , all active intervals have positive length, the active line at $x = 0$ is unique, and no inactive line touches a positive envelope vertex or the terminal vertex. Equivalently, the relevant intersection and dominance inequalities are strict. A one-line envelope with all other lines strictly inactive is included.

Lemma 2.2 (Envelope calculus). *The function I is convex, continuously differentiable on $(0, \infty)^d$, positively homogeneous of degree two, and quadratic on each strict chamber. Moreover,*

$$\partial_{a_i} I(a) = \ell_i(a), \quad \sum_i \ell_i = T, \quad \sum_i \lambda_i \ell_i = \max_i a_i, \quad \sum_i a_i \ell_i = 2I(a). \quad (2.5)$$

If the active list, from left to right, is $i_1, \dots, i_m$, then $\lambda_{i_1} > \dots > \lambda_{i_m}$ and

$$I(a) = \frac{a_{i_m}^2}{2\lambda_{i_m}} + \sum_{j=1}^{m-1} \frac{(a_{i_j} - a_{i_{j+1}})^2}{2(\lambda_{i_j} - \lambda_{i_{j+1}})}. \quad (2.6)$$

Proof. Convexity follows by integrating a maximum of affine functions of a . The substitution $x = cy$ gives $I(ca) = c^2 I(a)$ for $c > 0$. Distinct slopes imply that the maximizer is unique away from finitely many points. On a compact neighborhood of a , every positive envelope is supported in a common bounded interval. Dominated convergence applied to difference quotients therefore gives $\partial_{a_i} I = \ell_i$. The same argument applied to the maximizing indicators proves continuity of these derivatives, including at chamber walls.

The active intervals partition $[0, T]$, proving the first length identity. Since $h'_a = -\lambda_i$ on the i th active interval, integrating $-h'_a$ gives $\sum \lambda_i \ell_i = h_a(0) = \max a_i$. Euler's identity for a C^1 homogeneous function gives the final equality; alternatively integrate $h_a - xh'_a$.

As x increases, a line can replace the current maximum only if its slope $-\lambda_i$ is larger. Consecutive active crossings occur at

$$x_j = \frac{a_{i_j} - a_{i_{j+1}}}{\lambda_{i_j} - \lambda_{i_{j+1}}}, \quad x_0 = 0, \quad x_m = \frac{a_{i_m}}{\lambda_{i_m}}.$$

Integrating $a_{i_j} - \lambda_{i_j} x$ over $[x_{j-1}, x_j]$ and collecting the endpoint terms gives (2.6). $\square$

Formula (2.6) is to be used with the *actual* upper-envelope list. Summing the displayed crossing terms over an arbitrary ordering of all constraints is generally wrong.

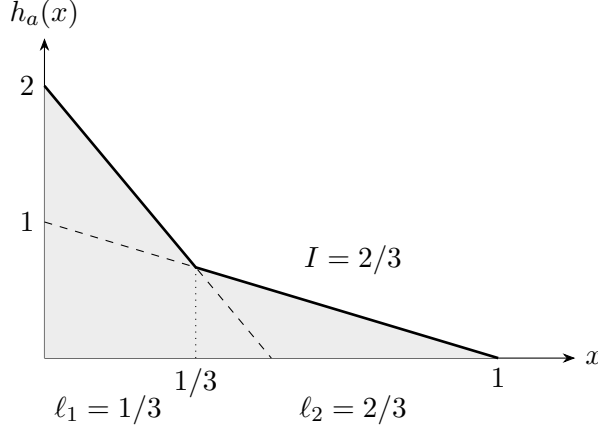


Figure 1: The envelope of $2 - 4x$, $1 - x$, and zero. Early digits pay the first constraint; later digits pay the second. The shaded area is the leading rate.

3 The joint logarithmic law, including continuum constraints

For this section only, assume the digit distribution function F_V satisfies

$$\frac{\log F_V(z)}{\log z} \longrightarrow \beta \in (0, \infty) \quad (z \downarrow 0). \quad (3.1)$$

No density assumption is needed here.

Theorem 3.1 (Upper-envelope small-deviation law). *Under (3.1),*

$$\log \mathbb{P}(E_t(a)) = -\beta I(a)t^2 + o(t^2). \quad (3.2)$$

If $\log F_V(z) = \beta \log z + O(1)$ uniformly for $0 < z \leq 1$, the error can be sharpened to $O(t \log t)$, locally uniformly in a .

The same statements hold with the finite set of constraints replaced by all $\lambda \in [B, A]$, $0 < B < A < \infty$, with continuous positive threshold profile $a(\lambda)$ and $w(\lambda) = 1 - e^{-\lambda}$; replace the maximum in (2.3) by the supremum over this interval.

Proof. Write

$$b_{i,n} = a_i t + \log w_i - \lambda_i n, \quad b_n = \max_i b_{i,n}.$$

If $E_t(a)$ occurs, positivity of every summand implies $V_n \leq \min(1, e^{-b_n})$ for each n . Independence gives the upper bound

$$\mathbb{P}(E_t(a)) \leq \prod_{n \geq 0} F_V(e^{-(b_n)_+}). \quad (3.3)$$

Only $O(t)$ factors differ from one. Since $\log w_i$ is bounded,

$$\sum_{n \geq 0} (b_n)_+ = t^2 I(a) + O(t). \quad (3.4)$$

This is a Riemann sum for $th_a(n/t)$; its support has length $O(t)$ and its normalized summand is uniformly Lipschitz.

For the converse, let $\lambda_{\min} = \min_i \lambda_i$ and choose

$$N = \left\lceil T(a)t + \frac{\log 2}{\lambda_{\min}} \right\rceil.$$

For every i the normalized deterministic tail is bounded by

$$e^{a_i t} w_i \sum_{n \geq N} e^{-\lambda_i n} = e^{a_i t - \lambda_i N} \leq \frac{1}{2}.$$

Impose the independent box conditions

$$V_n \leq c_n := \min \left(1, \frac{e^{-b_n}}{2N} \right), \quad 0 \leq n < N. \quad (3.5)$$

Each normalized core summand is at most $1/(2N)$, so these conditions imply $E_t(a)$. Thus

$$\mathbb{P}(E_t(a)) \geq \prod_{n < N} F_V(c_n), \quad \sum_{n < N} -\log c_n = t^2 I(a) + O(t \log t).$$

When $\log F_V(z) = \beta \log z + O(1)$, both product bounds give the claimed quantitative error. Under the weaker condition (3.1), for every $\varepsilon > 0$ there is a constant C_ε such that, for all $s \geq 0$,

$$|\log F_V(e^{-s}) + \beta s| \leq \varepsilon s + C_\varepsilon.$$

Apply this to both products, divide by t^2 , and then let $\varepsilon \downarrow 0$.

For the continuum case use $b_n = \sup_{\lambda \in [B, A]} (a(\lambda)t + \log w(\lambda) - \lambda n)$. The same deterministic implication, independence, and single common box remain valid. Compactness bounds all supports, slopes, and normalization constants. The resulting envelope is Lipschitz and decreasing until zero, so the Riemann estimate and the uniform tail cutoff hold without change. The event is measurable: the series defining X_λ is uniformly convergent on $[B, A]$, and continuous parameter constraints can be checked on a fixed countable dense subset. $\square$

Remark 3.2 (Why the maximum, not the sum?). Charging a separate independent cost for each parameter would produce $\sum_i a_i^2/(2\lambda_i)$. That double-counts a digit which is already small enough to satisfy several constraints. The correct leading cost is one maximum per digit, and hence the area of one upper envelope. The inequalities

$$\max_i \frac{a_i^2}{2\lambda_i} \leq I(a) \leq \sum_i \frac{a_i^2}{2\lambda_i} \quad (3.6)$$

are immediate, but their usually strict gap contains the dependence information.

4 A sharp three-scale expansion by independent simplex blocks

For the refined results assume that the digits have a density on $[0, 1]$ with

$$f(z) = cz^{\beta-1}(1 + O(z^\delta)), \quad c, \beta, \delta > 0, \quad z \downarrow 0. \quad (4.1)$$

This includes uniform digits ($\beta = c = 1$) and Beta($\beta, 1$) digits ($c = \beta$).

Theorem 4.1 (Sharp joint lower-tail expansion). *Suppose (4.1) holds and a is in a strict envelope chamber. Then, as $t \rightarrow \infty$,*

$$\log \mathbb{P}(E_t(a)) = -\beta I(a)t^2 - \beta T(a)t \log t + A_{\beta,c}(a)t + O((\log t)^2), \quad (4.2)$$

where

$$\begin{aligned} A_{\beta,c}(a) = & \beta T(a) - \frac{\beta}{2} \max_i a_i \\ & - \beta \sum_{\ell_i > 0} \ell_i \log(\beta w_i \ell_i) + T(a) \log(c\Gamma(\beta)). \end{aligned} \quad (4.3)$$

The error is uniform on compact subsets of a strict chamber. For uniform digits,

$$A(a) = T(a) - \frac{1}{2} \max_i a_i - \sum_{\ell_i > 0} \ell_i \log(w_i \ell_i). \quad (4.4)$$

The $t \log t$ term is an entropy cost for making a sum of order t positive coordinates small. The active block lengths tell us how many coordinates each constraint must control. The proof below also supplies finite computable bounds; it does not assume a multivariate saddle-point formula.

4.1 The simplex calculation and the refined lattice sum

Lemma 4.2 (Weighted Dirichlet simplex). *For $N \geq 0$, positive $A_1, \dots, A_N$, $b > 0$, and $\beta > 0$,*

$$\int_{\substack{u_j \geq 0 \\ \sum_{j=1}^N A_j u_j \leq b}} \prod_{j=1}^N c u_j^{\beta-1} du_j = \frac{(c\Gamma(\beta))^N b^{\beta N}}{\Gamma(\beta N + 1) \prod_{j=1}^N A_j^{\beta}}. \quad (4.5)$$

When $b \leq \min_j A_j$, this simplex lies in $[0, 1]^N$.

Proof. Substitute $y_j = A_j u_j / b$. The remaining Dirichlet integral is $\Gamma(\beta)^N / \Gamma(\beta N + 1)$, obtained by iterating the beta integral, or by introducing the slack coordinate $y_{N+1} = 1 - \sum_{j=1}^N y_j$ with exponent zero. The assertion about the cube follows from $u_j \leq b/A_j$. The empty integral is one. $\square$

Lemma 4.3 (Lattice cost). *Let $b_{i,n} = a_i t + \log w_i - \lambda_i n$ and $b_n = \max_i b_{i,n}$. In a fixed strict chamber,*

$$S(t) := \sum_{n \geq 0} (b_n)_+ = I(a)t^2 + \frac{1}{2} \max_i a_i t + t \sum_i \ell_i(a) \log w_i + O(1). \quad (4.6)$$

If N_i counts indices assigned to active line i , with only $O(\log t)$ indices near the crossings or the final zero omitted, then $N_i = t\ell_i + O(\log t)$ and

$$\sum_i \log \Gamma(\beta N_i + 1) = \beta T t \log t + \beta t \sum_{\ell_i > 0} \ell_i \log(\beta \ell_i) - \beta T t + O((\log t)^2). \quad (4.7)$$

Proof. The sum of a piecewise linear compactly supported function on a unit grid is its integral plus half its value at zero, with an error bounded by a constant times the sum of its finitely many slope jumps. One elementary proof uses the trapezoidal rule: it is exact on every grid cell not containing a breakpoint, and the error on a cell containing a breakpoint is bounded by its slope jump. Apply this to $x \mapsto t h_a(x/t)$, whose slopes are fixed. Thus

$$\sum_{n \geq 0} t h_a(n/t) = I(a)t^2 + \frac{1}{2} h_a(0)t + O(1).$$

Adding the fixed intercept shifts $\log w_i$ moves each crossing by $O(1)$ indices. Away from those finitely many indices it adds exactly $\log w_i$ on the i th active block. The affected indices have bounded differences between competing lines; their correction is $O(1)$. This proves (4.6).

The index counts follow from the interval lengths. Stirling's formula gives

$$\log \Gamma(\beta N + 1) = \beta N \log(\beta N) - \beta N + O(\log(N + 2)).$$

Changing $N = t\ell_i$ by $O(\log t)$ changes the right side by $O((\log t)^2)$ when $\ell_i > 0$. Summing over the fixed active list proves (4.7). Strict-chamber compactness bounds the nonzero lengths away from zero and the crossing slopes away from degeneracy. $\square$

4.2 The upper bound

Fix a constant $K > 2$ sufficiently large that $K\delta > 2$. Retain all indices with $b_n \geq K \log t$, and assign each to its maximizing line. Let B_i^+ be the resulting block and $N_i^+ = |B_i^+|$. The necessary conditions

$$\sum_{n \in B_i^+} e^{b_{i,n}} V_n \leq 1$$

use disjoint digits. Within each necessary simplex every coordinate is at most t^{-K} . Consequently (4.1) can be used uniformly: replacing the joint density there by $\prod c v_n^{\beta-1}$ changes the logarithm of its integral by $O(t^{1-K\delta})$. Extending the integration to the full nonnegative simplex is an upper bound, and theorem 4.2 gives

$$\begin{aligned} \log \mathbb{P}(E_t(a)) \leq & -\beta \sum_i \sum_{n \in B_i^+} b_{i,n} + \sum_i N_i^+ \log(c\Gamma(\beta)) \\ & - \sum_i \log \Gamma(\beta N_i^+ + 1) + O(1). \end{aligned} \quad (4.8)$$

Only $O(\log t)$ positive indices near the terminal zero are omitted, and their total cost is $O((\log t)^2)$. In particular the double sum in (4.8) is $S(t) + O((\log t)^2)$. Substituting theorem 4.3 proves the upper half of (4.2).

4.3 A matching lower bound and why transition strips matter

Set

$$\varepsilon = t^{-K}, \quad N = \left\lceil \max_i \frac{a_i t + K \log t}{\lambda_i} \right\rceil.$$

All normalized deterministic tails are at most ε . An index $n < N$ belongs to core B_i when

$$b_{i,n} \geq K \log t, \quad b_{i,n} - b_{j,n} \geq K \log t \quad (j \neq i). \quad (4.9)$$

Let S be the residual set. In a strict chamber there are only $M = |S| = O(\log t)$ residual indices: they lie in logarithmic-width strips around the finitely many crossings and around the final zero. Each active core has $|B_i| = t\ell_i + O(\log t)$.

On every core impose the simplex constraint

$$\sum_{n \in B_i} e^{b_{i,n}} V_n \leq b_t, \quad b_t = 1 - (d+2)\varepsilon,$$

and on residual indices impose

$$V_n \leq \min \left(1, \frac{e^{-b_n \varepsilon}}{\max(1, M)} \right). \quad (4.10)$$

For coordinate i , its own core contributes at most b_t , every other core contributes at most εb_t , all residual indices together contribute at most ε , and the tail contributes at most ε . Thus the total is less than one for all large t . Inactive coordinates have no own core and satisfy an even smaller bound. The constructed event is therefore contained in $E_t(a)$.

The core simplexes lie in $[0, t^{-K}]^{|B_i|}$, so the density estimate and theorem 4.2 give their probabilities with a bounded logarithmic error. The budget factor contributes $\beta \sum_i |B_i| \log b_t = O(t^{1-K})$. For residual indices, (4.1) implies the global comparison $\log F_V(z) = \beta \log z + O(1)$ on $(0, 1]$. Hence their log probability is

$$-\beta \sum_{n \in S} (b_n)_+ + O((\log t)^2).$$

The extra $K \log t + \log \max(1, M)$ cost per strip index is responsible for the stated remainder. Combining cores and strips restores $-\beta S(t)$, up to $O((\log t)^2)$. The total numbers of active core indices differ from $t\ell_i$ by only $O(\log t)$, so (4.7) applies. This proves the matching lower bound and completes theorem 4.1.

Remark 4.4 (A potentially serious proof error avoided). A strip near an *interior* crossing can contain $O(\log t)$ indices each with cost $\Theta(t)$. Its total base cost is not $O((\log t)^2)$. It would be wrong to discard those digits. The lower bound above pays their original βb_n costs explicitly and loses only the extra logarithmic safety factors. That is what makes the linear coefficient in (4.2) valid.

Corollary 4.5 (Scalar specialization). *For one parameter λ , $w = 1 - e^{-\lambda}$, and $s \rightarrow \infty$,*

$$\begin{aligned} -\log \mathbb{P}(X \leq e^{-s}) &= \frac{\beta s^2}{2\lambda} + \frac{\beta s}{\lambda} \log s - D_{\lambda, \beta, c} s + O((\log s)^2), \\ D_{\lambda, \beta, c} &= \frac{\beta}{\lambda} - \frac{\beta}{2} - \frac{\beta}{\lambda} \log \frac{\beta w}{\lambda} + \frac{1}{\lambda} \log(c\Gamma(\beta)). \end{aligned} \quad (4.11)$$

This scalar order of expansion is a consistency check and an input to marginal standardization, not a claim to supersede the repository's much finer one-parameter Fabius asymptotics. The $n = 0$ indexing and the factor $w = 1 - e^{-\lambda}$ both matter for the linear coefficient.

5 Marginal standardization and universal copula asymptotics

Write F_i for the marginal distribution function of X_i and Q_i for its inverse near zero. Under (4.1), F_i is continuous and strictly increasing sufficiently close to zero. Indeed, a sufficiently small positive tail has positive probability, and the first digit has a strictly positive density near zero; conditioning on that tail gives positive mass to each small open interval. The first digit also makes the law atomless. Thus the inverses used below are unambiguous for small arguments.

Define the lower copula by

$$C(u_1, \dots, u_d) = \mathbb{P}(X_i \leq Q_i(u_i) : 1 \leq i \leq d). \quad (5.1)$$

For a power-ray vector $\alpha \in (0, \infty)^d$ put

$$b_i = \sqrt{2\lambda_i \alpha_i}, \quad K(\alpha) = I(b), \quad H(\alpha) = \sum_{\ell_i(b) > 0} \ell_i(b) \log \frac{b_i/\lambda_i}{\ell_i(b)}. \quad (5.2)$$

The envelope and lengths here use the *canonical* intercepts b_i , without β .

Theorem 5.1 (Universal copula rate and first correction). *Assume (4.1). For every positive α ,*

$$\lim_{L \rightarrow \infty} \frac{\log C(e^{-\alpha_1 L}, \dots, e^{-\alpha_d L})}{L} = -K(\alpha). \quad (5.3)$$

If b lies in a strict envelope chamber, then

$$\log C(e^{-\alpha_1 L}, \dots, e^{-\alpha_d L}) = -K(\alpha)L + \sqrt{\beta L} H(\alpha) + O((\log L)^2). \quad (5.4)$$

In particular, the leading rate is independent of c and β ; H is independent of c and the weight factors w_i . Moreover,

$$\max_i \alpha_i \leq K(\alpha) \leq \sum_i \alpha_i, \quad H(\alpha) \geq 0. \quad (5.5)$$

In a strict chamber $H = 0$ exactly when a single line is active.

5.1 Quantile inversion with the needed accuracy

Let $t = \sqrt{L}$ and $\tilde{b}_i = b_i/\sqrt{\beta}$. Then

$$-\log Q_i(e^{-\alpha_i L}) = \tilde{b}_i t - \log t + d_i + O((\log t)^2/t), \quad (5.6)$$

where

$$d_i = 1 - \log \frac{w_i \beta \tilde{b}_i}{\lambda_i} + \frac{1}{\beta} \log(c\Gamma(\beta)) - \frac{\lambda_i}{2}. \quad (5.7)$$

To verify this, substitute $s = \tilde{b}_i t - \log t + d$ into (4.11). The quadratic term gives $\alpha_i t^2$ and a $-(\beta \tilde{b}_i/\lambda_i)t \log t$ contribution. The $s \log s$ term cancels the latter. The remaining coefficient of t vanishes precisely for (5.7); the residual is $O((\log t)^2)$. This substitution is also a rigorous inversion: changing s by $\pm M(\log t)^2/t$ changes the explicit main expression by $\pm(\beta \tilde{b}_i/\lambda_i)M(\log t)^2 + o((\log t)^2)$. Choose M larger than the uniform error constant and use monotonicity to bracket the true inverse. No differentiated remainder is assumed.

5.2 The cancellation which produces universality

Set

$$a_i(t) = \tilde{b}_i + \frac{-\log t + d_i}{t} + O((\log t)^2/t^2).$$

For a strict chamber these vectors remain in one compact chamber neighborhood. Applying theorem 4.1 uniformly gives

$$-\beta I(a(t))t^2 - \beta T(a(t))t \log t + A_{\beta,c}(a(t))t + O((\log t)^2).$$

By theorem 2.2 and piecewise quadraticity,

$$I(a(t)) = I(\tilde{b}) + \frac{1}{t} \sum_i \ell_i(\tilde{b})(-\log t + d_i) + O((\log t)^2/t^2).$$

All $t \log t$ terms cancel because $\sum \ell_i = T$. The coefficient of t is

$$A_{\beta,c}(\tilde{b}) - \beta \sum_i \ell_i(\tilde{b})d_i.$$

Insert (4.3) and (5.7). The terms containing $\log(c\Gamma(\beta))$, $\log w_i$, and the free term βT cancel. The remaining endpoint terms cancel by $\sum_i \lambda_i \ell_i(\tilde{b}) = \max_i \tilde{b}_i$. What remains is

$$\beta \sum_i \ell_i(\tilde{b}) \log \frac{\tilde{b}_i/\lambda_i}{\ell_i(\tilde{b})} = \sqrt{\beta} H(\alpha).$$

Homogeneity gives $\beta I(\tilde{b}) = I(b) = K(\alpha)$, proving (5.4).

For the leading formula at a chamber wall, the scalar leading equivalent in theorem 4.5 still gives $-\log Q_i(e^{-\alpha_i L})/\sqrt{L} \rightarrow \tilde{b}_i$. Use the locally uniform bound of theorem 3.1 and continuity of I . The bounds for K follow from (3.6). An active interval for line i lies in $[0, b_i/\lambda_i]$, so $\ell_i \leq b_i/\lambda_i$ and every summand in H is nonnegative. With two or more active lines, at least one such inclusion is strict; with only one it is an equality. This finishes the proof.

Remark 5.2 (Endpoint universality has a precise meaning). Universality here is invariance under replacing the uniform digit law by a law satisfying (4.1), *after its own marginals are standardized*. It does not assert that the unstandardized probabilities, whole copulas, covariances, or lower-order bounded oscillations coincide. The endpoint exponent β survives at the square-root scale.

6 From logarithmic rates to genuine regular variation

An expansion with an $O((\log L)^2)$ remainder does *not* by itself imply regular variation: that remainder could oscillate too fast under a fixed increment of L . To justify the standard residual-dependence terminology, a separate argument is required. We give one using convexity.

Assume in this section that (4.1) holds and the density

$$g(z) = e^{-z} f(e^{-z}) \mathbf{1}_{\{z \geq 0\}} \quad (6.1)$$

of $Z = -\log V$ is log-concave on $\mathbb{R}$. Both uniform and Beta($\beta, 1$) digits satisfy this hypothesis, since $g(z) = \beta e^{-\beta z} \mathbf{1}_{\{z \geq 0\}}$ in the latter case.

Lemma 6.1 (Convexity in log-threshold coordinates). *The function*

$$G(s) = -\log \mathbb{P}(X_i \leq e^{-s_i} : 1 \leq i \leq d) \quad (6.2)$$

is finite, convex, and coordinatewise nondecreasing on $\mathbb{R}^d$.

Proof. For the first N digits, the feasible set in (s, z) is

$$z_n \geq 0, \quad \sum_{n < N} \exp(s_i + \log w_i - \lambda_i n - z_n) \leq 1 \quad (1 \leq i \leq d).$$

It is convex, as each displayed exponential sum is convex. Its indicator is log-concave in the extended sense. Multiplying by the log-concave product density $\prod_{n < N} g(z_n)$ preserves log-concavity. Prékopa's marginal theorem [4] therefore makes the finite probability log-concave in s . These probabilities decrease to the infinite-series probability; the defining log-concavity inequality passes to the limit. The limit is strictly positive for every finite s by the box construction. Taking minus its logarithm proves convexity. Monotonicity is immediate from the events. $\square$

Lemma 6.2 (Asymptotic slopes of a convex rate). *Let $s_t/t \rightarrow a \in (0, \infty)^d$ and let p_t be any subgradient of G at s_t . Then*

$$\frac{p_t}{t} \rightarrow \beta \nabla I(a) = \beta(\ell_1(a), \dots, \ell_d(a)). \quad (6.3)$$

Proof. By the box estimate, $G(tx) = \beta t^2 I(x) + O(t \log t)$ uniformly for x in any compact subset of the positive orthant. Apply the subgradient inequality at s_t with increments $\pm \varepsilon t e_i$. After division by εt^2 , it bounds $p_{t,i}/t$ between the backward and forward difference quotients of βI at s_t/t , plus $O(\log t/(\varepsilon t))$. First let $t \rightarrow \infty$, then $\varepsilon \downarrow 0$, and use the continuous differentiability in theorem 2.2. A finite convex function on $\mathbb{R}^d$ has subgradients at every point, so the argument applies to all choices. $\square$

Theorem 6.3 (Power-ray regular variation). *Under the assumptions of this section, for every $\alpha_i > 0$ and $k > 0$,*

$$\lim_{u \downarrow 0} \frac{C((ku)^{\alpha_1}, \dots, (ku)^{\alpha_d})}{C(u^{\alpha_1}, \dots, u^{\alpha_d})} = k^{K(\alpha)}. \quad (6.4)$$

This includes chamber boundaries. Thus the copula restricted to any fixed power ray is regularly varying at zero with index $K(\alpha)$.

Proof. Put $s_i(L) = -\log Q_i(e^{-\alpha_i L})$ and $\tilde{b}_i = \sqrt{2\lambda_i \alpha_i / \beta}$. Scalar leading asymptotics give $s_i(L) \sim \tilde{b}_i \sqrt{L}$. For a fixed real r , the scalar version of theorem 6.2, together with the two subgradient bounds for an increment of a convex function, gives

$$s_i(L+r) - s_i(L) = \frac{r \tilde{b}_i}{2\sqrt{L}} + o(L^{-1/2}). \quad (6.5)$$

Here is the detail needed to avoid differentiating an asymptotic series. The exact scalar identity is $G_i(s_i(L)) = \alpha_i L$. Subgradients at both endpoints are asymptotic to $\beta \tilde{b}_i \sqrt{L} / \lambda_i$. Their sandwich first bounds the inverse increment by $O(L^{-1/2})$, and then evaluates it as $\alpha_i r \lambda_i / (\beta \tilde{b}_i \sqrt{L})$, which is (6.5). Negative r follows by reversing the endpoints.

Apply the multivariate subgradient sandwich to G at $s(L)$ and $s(L + r)$. By theorem 6.2 the two subgradients, divided by $\sqrt{L}$, converge to $\beta \ell(\tilde{b})$. Hence

$$G(s(L + r)) - G(s(L)) \longrightarrow \frac{r\beta}{2} \sum_i \tilde{b}_i \ell_i(\tilde{b}) = r\beta I(\tilde{b}) = rK(\alpha).$$

Take $r = -\log k$ and exponentiate the negative difference. □

In the weak lower-tail dependence notation of Tankov [6], the result gives

$$\chi(\alpha) = \frac{\max_i \alpha_i}{K(\alpha)}. \quad (6.6)$$

For the diagonal in two dimensions, genuine regular variation yields the Ledford–Tawn representation $C(u, u) = u^{1/\eta} \mathcal{L}(u)$ with $\mathcal{L}$ slowly varying at zero [5]. Thus $\eta = 1/K(1, 1)$ is justified in the usual sense, not merely defined by a logarithmic ratio.

Remark 6.4 (What is not implied). Power-ray regular variation is not a proof of a full multivariate hidden regular-variation measure, angular limit, or copula-density asymptotic. For general digits satisfying (4.1) but not log-digit log-concavity, theorem 5.1 still holds; theorem 6.3 is not asserted without an additional argument.

7 The complete bivariate phase diagram

Let the two decay rates be $A > B > 0$, and let the corresponding raw threshold intercepts be $a, b > 0$. The envelope consists of zero, $a - Ax$, and $b - Bx$.

Proposition 7.1 (Three phases). *The raw rate is*

$$I(a, b) = \begin{cases} b^2/(2B), & a \leq b, \\ b^2/(2B) + (a - b)^2/(2(A - B)), & b < a < (A/B)b, \\ a^2/(2A), & a \geq (A/B)b. \end{cases} \quad (7.1)$$

In the middle phase the active lengths are

$$\ell_1 = \frac{a - b}{A - B}, \quad \ell_2 = \frac{b}{B} - \frac{a - b}{A - B}. \quad (7.2)$$

For copula exponents $\alpha, \gamma > 0$, put $a = \sqrt{2A\alpha}$ and $b = \sqrt{2B\gamma}$. Then

$$K(\alpha, \gamma) = \begin{cases} \gamma, & \alpha \leq (B/A)\gamma, \\ \gamma + \frac{(\sqrt{A\alpha} - \sqrt{B\gamma})^2}{A - B}, & (B/A)\gamma < \alpha < (A/B)\gamma, \\ \alpha, & \alpha \geq (A/B)\gamma. \end{cases} \quad (7.3)$$

Proof. If $a \leq b$, the slower line begins higher and decreases more slowly, so it dominates. If $a/A \geq b/B$, the faster line remains higher until the other line reaches zero. In the remaining case they switch at $x_0 = (a - b)/(A - B)$ before the slower line reaches zero. Integrate the first line on $[0, x_0]$ and the second on $[x_0, b/B]$. Substitution of the canonical intercepts gives (7.3). □

The leading function is C^1 across both boundaries, as the general geometry predicts. Its second derivatives change there. In a one-line phase, an additional parameter constraint is free *at the leading logarithmic scale*; this does not mean the corresponding event is exactly redundant at finite thresholds.

Theorem 7.2 (Diagonal exponent and explicit correction). *For uniform digits and $A > B > 0$, put $L = \log(1/u)$. Then*

$$\log C(u, u) = -\kappa L + H\sqrt{L} + O((\log L)^2), \quad (7.4)$$

where

$$\kappa = \frac{2\sqrt{A}}{\sqrt{A} + \sqrt{B}}, \quad H = \sqrt{\frac{2}{B}} \log \left(1 + \sqrt{\frac{B}{A}} \right). \quad (7.5)$$

The lower-tail dependence coefficient is zero, while the residual coefficients are

$$\lambda_L := \lim_{u \downarrow 0} \frac{C(u, u)}{u} = 0, \quad \eta = \frac{1 + \sqrt{B/A}}{2}, \quad \bar{\chi} = 2\eta - 1 = \sqrt{B/A}. \quad (7.6)$$

For digits satisfying (4.1), the same logarithmic expansion holds with H replaced by $\sqrt{\beta}H$. The regular-variation interpretation of η additionally uses the assumption of theorem 6.3.

Proof. With $a = \sqrt{2A}$ and $b = \sqrt{2B}$, both lines are active and

$$\ell_1 = \frac{\sqrt{2}}{\sqrt{A} + \sqrt{B}}, \quad \ell_2 = \frac{\sqrt{2}\sqrt{A}}{\sqrt{B}(\sqrt{A} + \sqrt{B})}.$$

Formula (7.3) gives κ . Both ratios $(a/A)/\ell_1$ and $(b/B)/\ell_2$ equal $1 + \sqrt{B/A}$. Their length sum is $\sqrt{2/B}$, proving the displayed H . Because $1 < \kappa < 2$, (7.4) implies $C(u, u)/u \rightarrow 0$. The residual formulas follow from theorem 6.3 and the definitions in [5, 6]. $\square$

For uniform digits, replacing each digit by $1 - U_n$ preserves the joint law and replaces each X_i by $1 - X_i$. Therefore the same results hold at the common upper corner. More generally, if a digit density has endpoint powers β_0 at zero and β_1 at one, apply the lower corner theorem separately to V and $1 - V$. The leading standardized exponent is the same, but the square-root corrections use $\sqrt{\beta_0}$ and $\sqrt{\beta_1}$ respectively.

7.1 An explicit Fabius example

Take

$$q = \frac{1}{2}, \quad r = 2^{-1/4}, \quad A = \log 2, \quad B = \frac{1}{4} \log 2.$$

The first marginal CDF is the Fabius function. Here

$$\kappa = \frac{4}{3}, \quad \eta = \frac{3}{4}, \quad \bar{\chi} = \frac{1}{2}, \quad H = \sqrt{\frac{8}{\log 2}} \log \frac{3}{2} = 1.377481422290 \dots \quad (7.7)$$

Consequently the conditional extreme-threshold probability is

$$\begin{aligned} & \mathbb{P}(F_r(X_r) \leq u \mid F_q(X_q) \leq u) \\ &= u^{1/3} \exp \left\{ H \sqrt{\log(1/u)} + O((\log \log(1/u))^2) \right\}. \end{aligned} \quad (7.8)$$

This tends to zero despite the complete sharing of the underlying digits. The conditioning event is an inequality of probability u , not the zero-probability equality $X_q = x$.

7.2 A nonuniform coalescence warning

As $A \downarrow B$, the fixed-parameter exponent tends to one, consistent with identical coordinates. But the displayed fixed-chamber correction tends to $\sqrt{2/B} \log 2$, whereas at exact equality $C(u, u) = u$ has no such term. There is no contradiction: the proof removes strips whose width constants involve inverse slope gaps. Those constants diverge as $A - B \rightarrow 0$, and fixed-chamber asymptotics are not uniform in this limit. A simultaneous coalescence/extreme-value transition is a genuine further problem, not resolved by substituting $A = B$ into (7.5).

8 Many parameters, a continuum, and optimal placement

Theorem 8.1 (Finite diagonal exponent). *Order the parameters as $\lambda_1 > \lambda_2 > \dots > \lambda_d > 0$. For uniform digits, and also for every digit density satisfying (4.1),*

$$\lim_{u \downarrow 0} \frac{\log C(u, \dots, u)}{\log u} = \kappa_d = 1 + \sum_{j=1}^{d-1} \frac{\sqrt{\lambda_j} - \sqrt{\lambda_{j+1}}}{\sqrt{\lambda_j} + \sqrt{\lambda_{j+1}}}. \quad (8.1)$$

All canonical lines are active, so the sharp correction in theorem 5.1 also applies.

Proof. The intercepts are $\sqrt{2\lambda_i}$. Consecutive crossings are

$$x_j = \frac{\sqrt{2}}{\sqrt{\lambda_j} + \sqrt{\lambda_{j+1}}}, \quad x_0 = 0, \quad x_d = \sqrt{2/\lambda_d}.$$

These abscissae are strictly increasing. More explicitly, the graph of $\sqrt{2\lambda}$ is strictly concave in λ , so intermediate nodes lie strictly above chords connecting nonadjacent nodes. Hence successive intersections indeed describe the upper envelope; no line is skipped. Apply (2.6). The last-line triangle has area one, and each crossing term simplifies to the displayed adjacent ratio. $\square$

Definition 8.2 (Continuum diagonal probability). For fixed $0 < B < A$, let

$$\mathcal{C}_{[B,A]}(u) = \mathbb{P}(F_{e^{-\lambda}}(X_{e^{-\lambda}}) \leq u \text{ for every } \lambda \in [B, A]). \quad (8.2)$$

This is the probability of a common extreme-quantile event along the entire parameter curve, not a finite-dimensional copula with an informal infinite index.

Theorem 8.3 (Continuum exponent). *For digits satisfying (4.1),*

$$\boxed{\lim_{u \downarrow 0} \frac{\log \mathcal{C}_{[B,A]}(u)}{\log u} = \kappa_\infty = 1 + \frac{1}{4} \log \frac{A}{B}.} \quad (8.3)$$

For uniform digits the logarithm equals $-\kappa_\infty L + O(\sqrt{L} \log L)$, where $L = \log(1/u)$. No continuum analogue of the sharper $H\sqrt{L}$ term is asserted here.

Proof. The series $X_{e^{-\lambda}}$ converges uniformly and continuously in $\lambda \in [B, A]$. Its marginal CDFs are jointly continuous by coupling and atomlessness. The positive density near zero gives continuous inverse CDFs for sufficiently small arguments, uniformly on compact parameter intervals. The event in (8.2) can therefore be checked on a countable dense set.

The scalar box bounds, uniformly for $\lambda \in [B, A]$, imply

$$-\log Q_{e^{-\lambda}}(e^{-L}) = \sqrt{2\lambda L/\beta} + O(\log L).$$

This weaker uniform inversion follows already from $-\log F_{e^{-\lambda}}(e^{-s}) = \beta s^2/(2\lambda) + O(s \log s)$; no interchange of finite-dimensional sharp limits is involved. Apply the continuum version of theorem 3.1. Homogeneity cancels β , and the relevant canonical envelope is

$$h(x) = \sup_{\lambda \in [B, A]} \left(\sqrt{2\lambda} - \lambda x \right)_+ = \begin{cases} \sqrt{2A} - Ax, & 0 \leq x \leq (2A)^{-1/2}, \\ (2x)^{-1}, & (2A)^{-1/2} \leq x \leq (2B)^{-1/2}, \\ \sqrt{2B} - Bx, & (2B)^{-1/2} \leq x \leq \sqrt{2/B}, \\ 0, & x \geq \sqrt{2/B}. \end{cases} \quad (8.4)$$

Indeed, the unconstrained maximizer satisfies $\lambda = 1/(2x^2)$. The first and last linear pieces have areas $3/4$ and $1/4$; the middle piece has area $\frac{1}{4} \log(A/B)$. This proves the limit. The uniform box error and the $O(\log L)$ quantile error both contribute at most $O(\sqrt{L} \log L)$ to the logarithm, proving the stated quantitative bound. $\square$

Theorem 8.4 (Optimal finite placement and saturation). *Fix endpoints $\lambda_1 = A$, $\lambda_d = B$ and $d \geq 2$. Among distinct ordered internal parameters, κ_d is uniquely maximized by*

$$\lambda_j = A \exp \left(-\frac{j-1}{d-1} \log \frac{A}{B} \right), \quad 1 \leq j \leq d. \quad (8.5)$$

Writing $D = \log(A/B)$, the optimum and its saturation gap are

$$\kappa_d^{\max} = 1 + (d-1) \tanh \frac{D}{4(d-1)}, \quad (8.6)$$

$$\kappa_\infty - \kappa_d^{\max} = \frac{D^3}{192(d-1)^2} + O((d-1)^{-4}). \quad (8.7)$$

Inserting any genuinely new parameter between two existing parameters strictly increases the exponent.

Proof. Set $\Delta_j = \log \lambda_j - \log \lambda_{j+1} > 0$. Then $\sum \Delta_j = D$ and each adjacent term in (8.1) is $\tanh(\Delta_j/4)$. The function $\tanh x$ is strictly concave on $(0, \infty)$, so Jensen's inequality gives (8.6), with equality precisely for all Δ_j equal. Expand $\tanh x = x - x^3/3 + O(x^5)$ to obtain (8.7). Finally, for $r, s > 0$, $\tanh(r+s) = (\tanh r + \tanh s)/(1 + \tanh r \tanh s) < \tanh r + \tanh s$; splitting a gap therefore increases the exponent. $\square$

d	$\kappa_d^{\max}$	κ_∞	$(d-1)^2(\kappa_\infty - \kappa_d^{\max})$
2	1.600000000	1.693147181	0.093147181
3	1.666666667	1.693147181	0.105922056
5	1.686291501	1.693147181	0.109690873
9	1.691417870	1.693147181	0.110675888
17	1.692713880	1.693147181	0.110924946
33	1.693038794	1.693147181	0.110987388

The table uses $A/B = 16$. Its last column converges to $(\log 16)^3/192 = 0.111008\dots$. The design criterion is specifically maximization of the simultaneous extreme-quantile exponent. It is not a claim of universal optimality for estimation, interpolation, or ordinary mean-square prediction.

9 Why a Gaussian limit gets these extremes wrong

9.1 Pearson correlation and the true residual coefficient

For any nondegenerate iid digit law with finite variance, the covariance sum in (2.1) gives

$$\rho(q, r) = \text{Corr}(X_q, X_r) = \frac{\sqrt{(1-q^2)(1-r^2)}}{1-qr}. \quad (9.1)$$

This calculation also appears in the common-digit source [1]. For $q = e^{-A}$ and $r = e^{-B}$,

$$\rho(q, r) = \frac{\sqrt{\sinh A \sinh B}}{\sinh((A+B)/2)}. \quad (9.2)$$

Proposition 9.1 (Correlation overstates Gaussian residual dependence). *For $A > B > 0$,*

$$\rho(q, r) > \frac{2\sqrt{AB}}{A+B} > \sqrt{B/A} = \bar{\chi}_{\text{common digit}}. \quad (9.3)$$

A Gaussian copula with correlation $\rho(q, r)$ has the smaller diagonal exponent $2/(1 + \rho(q, r))$, and therefore larger joint lower-tail probabilities at the leading exponent scale than the common-digit copula.

Proof. The function $f(x) = \log(\sinh x/x)$ is strictly convex on $(0, \infty)$, since

$$f''(x) = \frac{1}{x^2} - \frac{1}{\sinh^2 x} > 0.$$

Its strict midpoint inequality inserted into (9.2) yields the first inequality. The second follows from $2A/(A+B) > 1$.

For completeness, if two standard normal variables have correlation $\rho \in (0, 1)$, their joint lower tail at $-z$ has logarithm $-z^2/(1 + \rho) + O(\log z)$. The upper bound follows by applying a normal tail bound to their sum; a lower bound follows by integrating the joint density over the square $[-z - z^{-1}, -z]^2$. Since $\log \Phi(-z) = -z^2/2 + O(\log z)$, their copula exponent is $2/(1 + \rho)$, in agreement with the general Gaussian calculation in [6]. Compare this with (7.5). $\square$

9.2 Fixed correlation, a Gaussian limit, and two different answers

Let $v_2 > v_1$ and $\delta = v_2 - v_1 > 0$. For sufficiently large T set

$$q_1(T) = \tanh(T + v_1), \quad q_2(T) = \tanh(T + v_2). \quad (9.4)$$

Denote their common-digit copula by C_T .

Theorem 9.2 (Gaussian and extreme-tail limits do not commute). *For uniform digits the correlation in (9.4) is exactly $\rho = \text{sech } \delta$, independently of T , and C_T converges pointwise on $(0, 1)^2$ to the Gaussian copula C_ρ^G . Nevertheless,*

$$\lim_{T \rightarrow \infty} \lim_{u \downarrow 0} \frac{\log C_T(u, u)}{\log u} = \frac{2}{1 + e^{-\delta}}, \quad (9.5)$$

$$\lim_{u \downarrow 0} \lim_{T \rightarrow \infty} \frac{\log C_T(u, u)}{\log u} = \frac{2}{1 + \text{sech } \delta}. \quad (9.6)$$

The first value is strictly larger than the second.

Proof. The elementary hyperbolic subtraction formula transforms (9.1) into $\operatorname{sech}(v_2 - v_1)$, proving the exact correlation assertion. The Gaussianization itself is present in [1]; we give a short independent proof of precisely the finite-dimensional consequence needed here.

The centered standardized variable is

$$Y_q = \sqrt{12(1 - q^2)} \sum_{n \geq 0} q^n (U_n - \tfrac{1}{2}).$$

For any fixed real coefficients c_1, c_2 , the summands of $c_1 Y_{q_1(T)} + c_2 Y_{q_2(T)}$ are independent, centered, and bounded in absolute value by a deterministic number tending to zero. Their variance sum is the constant $c_1^2 + c_2^2 + 2\rho c_1 c_2$. The sum of their third absolute moments is at most their maximum absolute bound times that variance sum, and hence tends to zero. Taylor expansion of the characteristic functions, or the Lindeberg argument, proves a centered normal limit with that variance. Cramér–Wold gives the bivariate normal limit. Since the limiting marginal CDFs are continuous and strictly increasing, the marginal quantiles converge at each fixed $u \in (0, 1)$; the joint convergence then implies $C_T(u, v) \rightarrow C_\rho^G(u, v)$.

For each fixed T , write $\lambda_i(T) = -\log q_i(T)$. The exact common-digit exponent is

$$\kappa_T = \frac{2}{1 + \sqrt{\lambda_2(T)/\lambda_1(T)}}.$$

Since $-\log \tanh x \sim 2e^{-2x}$, $\sqrt{\lambda_2(T)/\lambda_1(T)} \rightarrow e^{-\delta}$. This proves (9.5). Taking $T \rightarrow \infty$ first gives the Gaussian copula, whose exponent was calculated in the preceding proof; this is (9.6). Finally $\operatorname{sech} \delta > e^{-\delta}$ for $\delta > 0$. $\square$

Example 9.3 (A numerical separation with exact constants). If $\delta = \log 2$, Pearson correlation is $\operatorname{sech}(\log 2) = 4/5$ for *every* T . The extreme-tail-first limit is $4/3$, whereas the Gaussian-first limit is $10/9$. Covariance does not merely converge to the wrong predictor of endpoint behavior: it is unchanged throughout the entire transition.

The distinction is structural. Gaussianization describes bounded standardized fluctuations when the largest digit weight vanishes. An endpoint event constrains an increasing number of positive digits simultaneously. Its logarithmic cost depends on which geometric decay rate is active at each digit index, information not determined by covariance.

10 Finite probability certificates and reproducible checks

The preceding asymptotic proof has a finite, independently checkable core. This is useful both for testing coefficients and for a future formalization. The following certificate does not require equal-rate thresholds or any asymptotic regime.

Proposition 10.1 (Finite block certificate). *Fix thresholds $x_i > 0$ and an integer $N \geq 1$, and put*

$$A_{i,n} = \frac{w_i q_i^n}{x_i} \quad (0 \leq n < N), \quad \tau_i = \frac{q_i^N}{x_i}.$$

Partition $\{0, \dots, N-1\}$ into disjoint cores $B_1, \dots, B_d$ and a residual set S . Choose positive core budgets b_j and caps $0 < c_n \leq 1$ for $n \in S$. For a nonempty core define $R_{ij} = \max_{n \in B_j} A_{i,n}/A_{j,n}$, and set $R_{ij} = 0$ for an empty core. Suppose

$$b_j \leq \min_{n \in B_j} A_{j,n} \quad (B_j \neq \emptyset), \quad (10.1)$$

$$\sum_j R_{ij} b_j + \sum_{n \in S} A_{i,n} c_n + \tau_i \leq 1 \quad (1 \leq i \leq d). \quad (10.2)$$

For $\text{Beta}(\beta, 1)$ digits,

$$\mathbb{P}(X_i \leq x_i \forall i) \geq \prod_j \frac{\Gamma(\beta + 1)^{|B_j|} b_j^{\beta|B_j|}}{\Gamma(\beta|B_j| + 1) \prod_{n \in B_j} A_{j,n}^\beta} \prod_{n \in S} c_n^\beta. \quad (10.3)$$

For any disjoint assigned blocks D_j of finitely many digit indices, an upper bound is

$$\mathbb{P}(X_i \leq x_i \forall i) \leq \prod_j \min \left(1, \frac{\Gamma(\beta + 1)^{|D_j|}}{\Gamma(\beta|D_j| + 1) \prod_{n \in D_j} A_{j,n}^\beta} \right). \quad (10.4)$$

For uniform digits, rational inputs make all quantities in these certificates rational.

Proof. On core B_j impose $\sum_{n \in B_j} A_{j,n} V_n \leq b_j$ and on S impose $V_n \leq c_n$. The row inequality (10.2) proves that the whole normalized sum in each coordinate is at most one, including its bounded infinite tail. The cube condition lets theorem 4.2 evaluate each core probability exactly, and independence multiplies the disjoint factors. For the upper bound, the full event implies $\sum_{n \in D_j} A_{j,n} V_n \leq 1$ for each assigned block. Extend the integration from the cube to the positive orthant, apply the same simplex formula, and use independence. Intersecting with the trivial probability bound one only improves the result. $\square$

10.1 What the included program actually checks

The standard-library program `verify.py` performs 804 exact rational envelope tests: randomized active-list identities, the full bivariate phase formula on a rational grid, and finite diagonal formulas. Its random seed is fixed. It also constructs an exact rational lower and upper probability certificate for

$$(q_1, q_2) = (1/2, 3/4), \quad (x_1, x_2) = (2^{-20}, 2^{-15}).$$

The certificate has $N = 46$, core sizes $(4, 4)$, and 38 residual digits. Every row budget and cube inclusion is checked with Python `Fraction` arithmetic; no floating-point sign decision is used for those assertions. Decimal evaluations of the resulting exact bounds are

$$-392.442411713 \dots \leq \log \mathbb{P}(X_1 \leq 2^{-20}, X_2 \leq 2^{-15}) \leq -231.988498892 \dots$$

These are deliberately simple certificates, not tight probability estimates. Their decimal representations are ordinary floating-point evaluations, not outward-rounded interval endpoints.

A separate numerical calculation checks the second-order copula cancellation for 100 choices of β, c . It also generates the following asymptotic table for uniform digits with

$$(\lambda_1, \lambda_2) = (4, 1), \quad (a_1, a_2) = (2, 1).$$

Here $I = 2/3$, $T = 1$, $\ell = (1/3, 2/3)$, and $A(a) = 0.948459414 \dots$. The lower and upper columns show

$$\frac{\log(\text{certificate bound}) + It^2 + Tt \log t}{t}.$$

t	Lower certificate	$A(a)$	Upper certificate
100	-1.125052	0.948459	0.889717
500	0.276169	0.948459	0.933357
2,000	0.708024	0.948459	0.943991
10,000	0.882048	0.948459	0.947413
50,000	0.930301	0.948459	0.948216

Both certificates approach the predicted coefficient. The coefficient itself need not lie between the two finite- t entries, because the normalized probability has its own nonzero remainder. Unlike the rational certificate, this large- t table is evaluated using ordinary floating-point logarithms and gamma functions. It illustrates a proved asymptotic mechanism; it is not an interval-certified experiment.

10.2 Reproduction and trust boundary

Run `python3 verify.py` in the package directory. It regenerates `verification.txt`, the CSV table, the exact certificate metadata, and the two LaTeX tables under `generated/`. Then run `pdflatex article.tex` twice to rebuild this PDF. Only the Python standard library is required for the checks. A standard TeX installation with Libertinus, TikZ, and the listed document packages is required for the PDF.

The exact checks verify finite arithmetic statements selected by the program. Python itself is not a proof-assistant kernel, and no claim is made that the program certifies the measure-theoretic passage to an infinite series, Prékopa's theorem, or the asymptotic errors. Those are proved in the text. Conversely, no plot or numerical trend is used as a hypothesis in those proofs.

11 Further research questions and a formalization roadmap

The following problems are not claimed solved. They are ordered roughly from a close extension of the present methods to a substantially broader program.

Research question 11.1 (Bounded terms and joint periodic phases). Can (4.2) be continued through the $(\log t)^2$, $\log t$, constant, and bounded oscillatory scales? The geometric lattice suggests several phases, including the crossing indices and the terminal cutoff. Determine when a genuine periodic or quasiperiodic function occurs, and distinguish lattice effects from the errors introduced by the strip construction. A multidimensional Laplace inversion would need a proved uniform remainder, not just a formal saddle expansion.

Research question 11.2 (Conditioning on an exact coordinate). Determine asymptotics of the joint density, conditional density, and $\mathbb{P}(X_r \leq y \mid X_q = x)$ in a specified endpoint scaling. This would approach the conditional-event problem suggested by the source report more directly than (7.8). Differentiating the current probability expansion is not justified and cannot be used as a shortcut.

Research question 11.3 (Coalescing parameters). Find a uniform transition law when $A - B \downarrow 0$ simultaneously with $u \downarrow 0$. The mismatch of the fixed-chamber correction at exact coalescence shows that a new scaling analysis is necessary. Identify the correct crossover variable rather than assuming that fixed-parameter asymptotics remain uniform.

Research question 11.4 (A simultaneous Gaussian/extreme limit). In theorem 9.2, let $L = \log(1/u)$ grow together with T . Which regimes interpolate between $2/(1 + \operatorname{sech} \delta)$ and $2/(1 + e^{-\delta})$? Because an individual largest standardized weight is of order e^{-T} , the quantity $e^{-T} \sqrt{L}$ is a natural candidate for locating a transition, but this is a heuristic scale, not a theorem. A uniform moderate-deviation analysis could test and refine it.

Research question 11.5 (A continuum second-order law). Compute the first correction to (8.3). The middle branch of (8.4) is curved and has continuously varying active parameters, so there is no fixed finite set of simplex blocks. It is not legitimate to send $d \rightarrow \infty$ in the finite H formula without controlling the growth of its error. Determine whether a $\sqrt{L} \log L$ term survives and how it depends on the endpoint digit index.

Research question 11.6 (Growing dimension and near-optimal meshes). How fast may $d = d(L)$ grow while the finite-dimensional sharp expansion remains uniform? Combine the gap in (8.7) with a uniform bound on transition-strip losses. The rate formula gives a deterministic design criterion; a growing-dimensional probability theorem would quantify when that criterion predicts actual finite-threshold performance.

Research question 11.7 (Regular variation without log-digit log-concavity). Does theorem 6.3 hold for every density satisfying (4.1)? The explicit logarithmic expansion alone is insufficient. One

possible route is a local ratio theorem for the core simplexes which controls probability changes under $O(L^{-1/2})$ log-threshold increments and treats all transition strips uniformly.

Research question 11.8 (Hidden regular variation and angular laws). Construct, when it exists, a full hidden limit measure beyond the power-ray limits. Identify its natural scaling and angular component, and relate changes of active envelope to changes in the limiting conditional digit profile. The piecewise-quadratic K provides a candidate rate geometry, not the measure itself.

Research question 11.9 (Dependent digits). Extend the envelope law to stationary Markov or mixing digits whose small-value probabilities have controlled conditional powers. Determine whether the leading cost is still an integral of a scalar envelope or becomes a matrix or entropy-rate variational problem. The present proof uses independence essentially in both its box and simplex steps.

Research question 11.10 (Signed sums and genuine parameter jets). Parameter derivatives of X_q introduce signed polynomial-geometric coefficients. The one-digit necessary bounds used here then fail because of cancellation. Develop a replacement for joint lower-corner events or small norm balls of actual jets. Results for a positive scalar polynomial-geometric series do not automatically solve this signed multivariate problem.

Research question 11.11 (Distinguishing covariance from endpoint structure). Can the exact relation $\bar{\chi} = \sqrt{B/A}$ be used to identify relative geometric decay rates from extremes, and how does that compare with covariance-based identification? Develop estimators and quantify bias caused by the explicit $H\sqrt{L}$ correction. The latter can be large at practical thresholds even when the limiting exponent is simple.

11.1 A staged Lean/Rocq route

The first useful formal target is [theorem 10.1](#), not the entire analytic manuscript. Encode rational weights, a partition of finite digit indices, row budgets, and the simplex expression. A proved checker can then turn an external proposed partition into a finite theorem, with a transparent boundary between arithmetic verification and probability semantics. The existing repository's emphasis on certificates makes this a natural integration point.

A second stage formalizes finite upper envelopes: active-list correctness, intersection ordering, area, and (2.5). These statements require only finite algebra and elementary integration. The bivariate phase formulas and optimal-mesh inequalities can then be checked independently of the full probability construction.

The analytic stages are more demanding: uniform convergence of the random series, independent-box products, the Dirichlet simplex integral, the transition-strip estimate, and local-uniform error bounds. Copula standardization should follow only after a quantitative inverse-bracketing lemma is available. The regular-variation route additionally needs a suitable Prékopa theorem or a specialized proof for exponential digits. This roadmap is a proposal; no unavailable library theorem or uncompiled Lean code is presented as completed work.

12 Conclusion

The common-digit construction has two distinct geometries. Its central fluctuations are governed by a covariance kernel and can become Gaussian. Its simultaneous endpoint events are governed by an upper envelope of geometric decay lines and by the volumes of independent simplex blocks. The latter geometry yields explicit joint rates, first corrections, residual coefficients, finite and continuum sampling formulas, and a constant-correlation example of noncommuting Gaussian and extreme limits.

These are substantial answers to the endpoint-dependence program attached to the repository’s common-digit Fabius family. They do not replace the remaining boundary-transseries, conditional-density, uniform-coalescence, or formal-verification tasks. The package separates the proved statements, classical inputs, finite computational checks, and unresolved questions so that each next step has a precise mathematical target.

A Source audit and reproducibility manifest

The mathematical baseline was the common-digit report at the pinned commit, not an unpinned description of the repository. The repository root README, Fabius README, and draft manifest were used to locate that report. Its sections on the common-digit construction, copula support, Gaussianization, and open endpoint laws were the relevant comparison points. The large repository was not exhaustively audited, and no claim is made that every historical draft or unpublished parallel contribution was inspected.

Targeted external checks covered the original Fabius construction, Prékopa’s marginal theorem, residual dependence, and weak copula tails. They establish appropriate classical attribution, not exhaustive novelty. The source report’s scalar small-ball conjecture already has an editorial correction in the inspected snapshot. We neither rely on its uncorrected form nor describe the scalar specialization here as a new resolution of that issue.

Package item	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Complete manuscript and compiled reading copy.
<code>verify.py</code>	Exact finite checks, probability certificate, and numerical tables; standard library only.
<code>verification.txt</code>	Output of the included program from this build.
<code>generated/</code>	Regenerated tables, CSV data, and exact-certificate input/budget metadata.
<code>README.md</code>	Build instructions and statement of proof status.
<code>SOURCE_AUDIT.md</code>	Pinned source locations, attribution, and novelty limitations.

References

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[Pinned source on GitHub](#). The file identifies an AI-generated research-report author; the repository owner is listed here for repository attribution, not as an inferred claim of sole manuscript authorship.
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